

Sound transmission through lightweight parallel plates. Part II: structure-borne sound

R.J.M. Craik*, R.S. Smith

*Department of Building Engineering and Surveying, Heriot-Watt University, Riccarton,
Edinburgh, EH14 4AS, UK*

Received 13 May 1999; received in revised form 17 November 1999; accepted 26 November 1999

Abstract

In Part I of this paper sound transmission through double leaf lightweight partitions was examined. It was shown that an important part of the overall transmission is determined by structure-borne sound transmission between the two leaves of the wall. In this report, Part II of the paper, structural coupling is examined in more detail. Two theories are presented. One is appropriate where the connection behaves as a series of points and the other where the connection behaves as a continuous line. It was found that an appropriate transition frequency between these two theories was where a half bending wavelength on the plate fitted between the nails or screws that form the coupling. In the most common forms of construction the connection will behave as a series of independent points for most of the frequency range. The continuous line model included the frame either as a beam or as a short plate. It was found that the agreement with the experimental data was better when the frame was modelled as a plate. There was good agreement between the measured and predicted data for transmission between the two leaves of the wall for a wide variety of structures. These theories can be incorporated into a statistical energy analysis model to enable the sound transmission throughout the entire structure to be predicted as was shown in Part I of this paper. © 2000 Elsevier Science Ltd. All rights reserved.

1. Introduction

Lightweight partitions, like that shown in Fig. 1, are widely used as internal partitions and are increasingly being used as party walls between dwellings. Part I [1] of this two part paper discussed the overall sound transmission through these double wall structures and in particular examined the airborne sound transmission paths.

* Corresponding author. Tel.: +44-131-449-5111; fax: +44-131-451-3161.

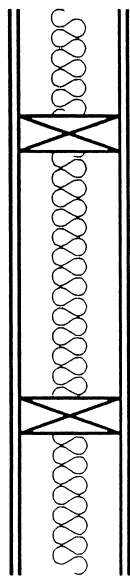


Fig. 1. Section through a lightweight double wall with a common frame providing structural connections between the two leaves of the wall.

This part (Part II) examines in more detail structure-borne sound transmission through the structural frame of these lightweight double walls.

Lightweight double walls are usually composed of a structural frame, which is made of a timber stud or metal channel, supporting plasterboard sheets (plates) on either side connected by a row of nail or screw fixings. The frame members are spaced from 300 to 600 mm centres depending upon the design requirements. The plasterboard plates are connected to the frames by nails or screws at spacings that vary from 150 to 600 mm. BS5234 [2] recommends 300 mm spacing for screws and 150 mm spacing for nails in a partition.

A major factor affecting the behaviour of these walls is the spacing between the nails or screws. Where the spacing is small the connection between each plate and the frame can be considered to be a continuous line. Where the spacing is large each point can be considered to be independent and so the coupling may be modelled as a series of independent point connections. It was found that an appropriate transition frequency between these two forms of construction was where a half bending wavelength on the plate fitted between the nails or screws.

Structure-borne sound transmission between point connected structures has been studied by several authors. Cremer et al. [3] describe the basic theory relating to point acting sound bridges and use the impedances of the components to compute power flow across a connection. Craik and Wilson [4] used these models to calculate of the coupling between the two leaves of a masonry cavity wall.

Structure-borne sound transmission for line connected plates has also been examined by many authors. In this paper the calculations for a joint with a beam are effectively

the same as those developed by Langley and Heron [5]. In the second model, where the connecting element is a plate, the theory resembles the model developed by Bhattacharya et al. [6]. The models are both developments of a simple rigid model in which the properties of the frame were neglected [7].

One of the limitations of the models developed in this paper is that the frame is assumed to be acoustically thin. Thin plate and beam theory has been assumed throughout even though in some cases the frequencies shown are above the arbitrary limit where $\lambda = 6h$.

Recently, a model based on a wave approach has been developed by Bosmans and Vermeir [8] for point connected plates. Although the model is computational intensive making it at present unsuitable for regular applications, it is the only model that can cover the transition region from point to line connections.

2. Theory

2.1. Theory of point connections

In a lightweight building the simplest case of coupling between the plasterboard and the frame occurs if the nails on each side are well separated so that they can be modelled as independent points. The model is further simplified if the nails on one side are assumed to be independent of the nails on the other side so that the coupling between each plate and the beam can be treated independently. The total coupling between the beam and either plate is then the coupling due to a single connection multiplied by the total number of connections.

This situation is shown in Fig. 2 which shows a plate coupled to a beam at a single point. If it is assumed that the velocity on plate 1 far from the connection is v_1 and

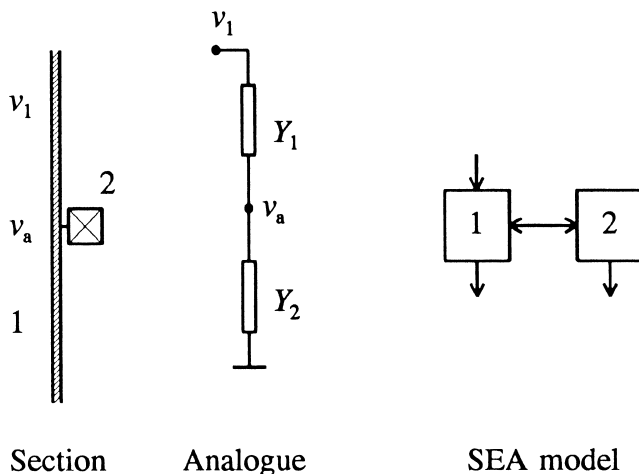


Fig. 2. A plate coupled to a beam at a point can be represented by two mobilities in series permitting the coupling between the two subsystems to be computed.

the velocity at the point of coupling is v_a then the system can be modelled as two mobilities in series as shown in the figure. If it is further assumed that the coupling is only due to forces and that no moment coupling is involved then the mobilities are simple point mobilities. This is valid for points far from plate edges and gives good agreement with experimental data for all conditions. The coupling loss factor for coupling between two subsystems, η_{12} , for r point connections can be given as [9],

$$\eta_{12} = \frac{r \operatorname{Re}(Y_2)}{\omega m_1 |Y_1 + Y_2|^2} \quad (1)$$

where Y is the mobility (or inverse of impedance) and m is the subsystem mass of the respective subsystems. The coupling loss factor in Eq. (1) may be used both for coupling from plate to beam and for beam to plate. The point mobilities of the plate and beam are given by [3],

$$Y_p = \frac{1}{8\sqrt{B_p \rho_s}}, \quad Y_b = \frac{1}{2\rho_l \left[\frac{B_b \omega^2}{\rho_l} \right]^{1/4} (1+i)} \quad (2)$$

where ρ_s is the surface density of the plate and ρ_l is the mass per unit length of the beam. For a plate B_p is the bending stiffness per unit width and is given by $Eh^3/12(1-\mu^2)$ where E is Young's Modulus, h is the thickness and μ is Poisson's ratio. For a beam B_b is the bending stiffness and is given by $Ebd^3/12$ where b is the beam width and d is the beam depth.

Although the nails may be well separated on any one side of the wall the nails may be arranged to be opposite each other on opposite sides of the wall as shown in Fig. 3.

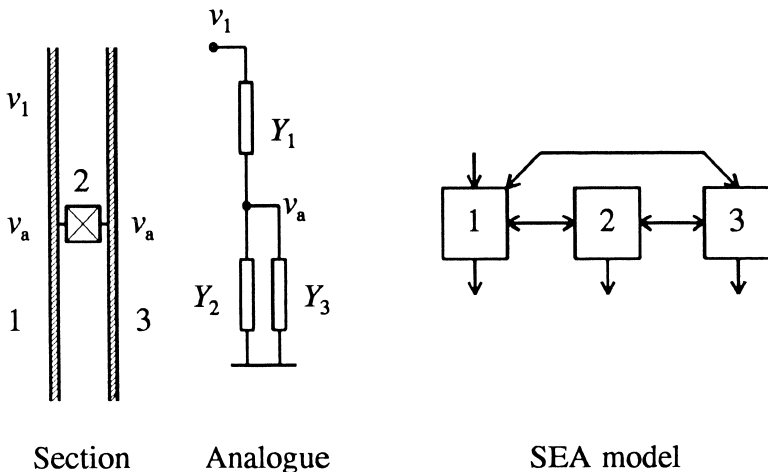


Fig. 3. Two plates connected to a beam by nails placed directly opposite each other can be represented by series and parallel mobilities from which the coupling may be found.

In this case a different analogue will apply which is also shown in Fig. 3. The beam is assumed to have a rigid cross section so that the velocity of the two opposite sides will be equal (v_a). At the point connection, the two plates and the beam will all have the same velocity and so coupling will take place between the source plate and beam as before but also directly between the two plates as shown in the SEA model.

The combined effective mobility of the beam and plate, subsystems 2 and 3, where Y_2 and Y_3 are the mobility of the beam and receiving plate, can be given by,

$$\frac{1}{Y_e} = \frac{1}{Y_2} + \frac{1}{Y_3} \quad (3)$$

which gives the coupling between the two plates, subsystems 1 and 3, as,

$$\eta_{13} = \frac{r}{m_1 \omega} \operatorname{Re} \left(\frac{1}{Y_3} \right) \frac{|Y_e|^2}{|Y_1 + Y_e|^2} \quad (4)$$

This equation can be used for any of the six coupling loss factors associated with the joint by interchanging the subscripts.

2.2. Theory for line connection

When the connection points are close together or the joint is formed from a continuous connection (e.g. glue) then it is not appropriate to use a model based on point connections and a model based on the interaction of semi-infinite plates can be used. The plates are assumed to be semi infinite and to intersect along an infinite joint (or two parallel joints). A wave of known amplitude (either a bending, longitudinal or transverse) is assumed incident on the joint and from the boundary conditions the amplitude of each wave leaving the joint can be found. This basic procedure is widely used and so full details of the calculations are not given.

There are three models presented in this paper for predicting sound transmission across a line connection between parallel plates using this basic procedure. The first model assumes that the frame connecting the plates is a beam. The second model assumes that the frame may be modelled as a finite width plate supporting full bending, longitudinal and transverse wave motion. In this case there are two parallel junctions one for each edge of the finite width plate. In the third model the frame is modelled as an SEA subsystem (a plate) and the two junctions are considered independently as two T junctions. Existing theories can then be used for these joints.

The coordinate system used is shown in Figs. 4 and 5 for the case of an interconnecting beam and plate, respectively.

The equation of motion for the bending waves on a thin plate may be written as [3],

$$\frac{B}{\rho_s} \nabla^4 \eta + \frac{\partial^2 \eta}{\partial t^2} = 0 \quad (5)$$

where η is the transverse displacement.

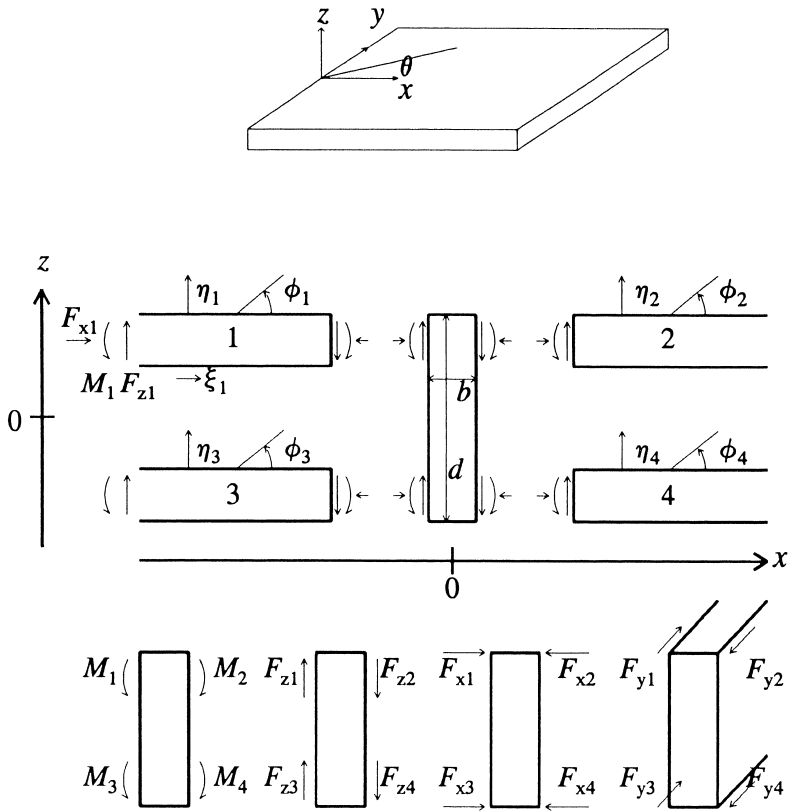


Fig. 4. Coordinate system used for a joint where the frame is modelled as a beam.

Similarly the in-plane waves must satisfy the equations[5],

$$\frac{E}{(1-\mu^2)} \left(\frac{\partial^2 \xi}{\partial x^2} + \mu \frac{\partial^2 \zeta}{\partial x \partial y} \right) + \frac{E}{2(1+\mu)} \left(\frac{\partial^2 \xi}{\partial y^2} + \frac{\partial^2 \zeta}{\partial x \partial y} \right) - \rho \frac{\partial^2 \xi}{\partial t^2} = 0 \quad (6)$$

and

$$\frac{E}{(1-\mu^2)} \left(\frac{\partial^2 \zeta}{\partial y^2} + \mu \frac{\partial^2 \xi}{\partial x \partial y} \right) + \frac{E}{2(1+\mu)} \left(\frac{\partial^2 \zeta}{\partial x^2} + \frac{\partial^2 \xi}{\partial x \partial y} \right) - \rho \frac{\partial^2 \zeta}{\partial t^2} = 0 \quad (7)$$

A bending travelling wave on plate 1 travelling towards the junction in a positive x and y direction can be given by,

$$\eta_0 = T_{B0} e^{-ik_{B0} \cos \theta_{B0} x} e^{-ik_{B0} \sin \theta_{B0} y} e^{i\omega t} \quad (8)$$

where T_{B0} represents the incident wave amplitude and θ_{B0} is the angle at which the wave is travelling (relative to the normal). All secondary waves produced by this

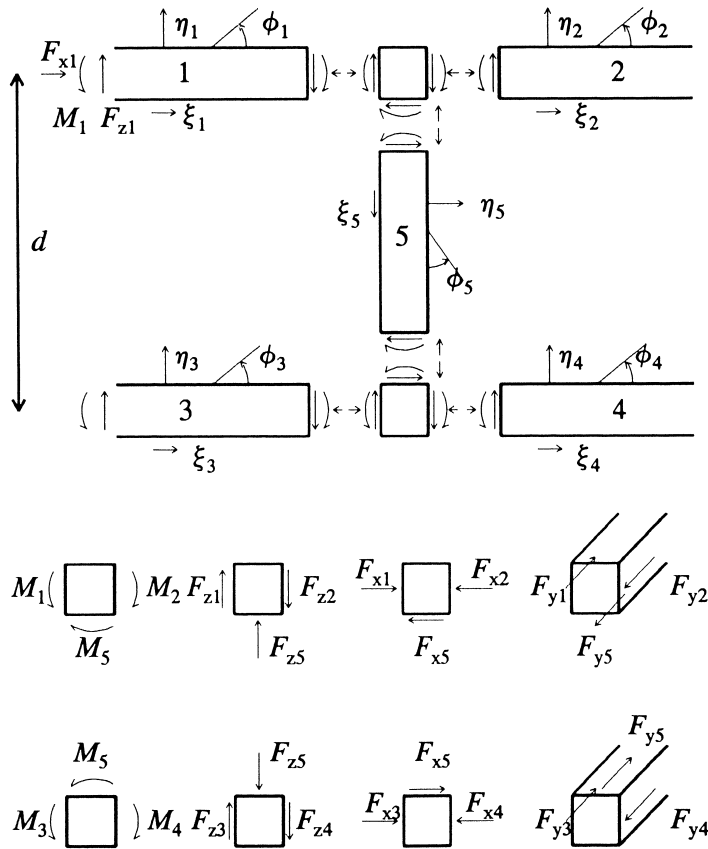


Fig. 5. Coordinate system used for a joint where the frame is modelled as a plate.

primary wave will have the same spacial dependance in the y direction and the same time dependance. For simplicity the terms relating to y and t are not included hereafter.

The total bending displacement on plate 1, η_1 , will be composed of a reflected wave, with amplitude T_{B1} , and a bending nearfield wave with amplitude T_{n1} , together with the incident wave (if the incident wave is a bending wave), to give,

$$\eta_1 = T_{B0} e^{-ik_{B0} \cos \theta_{B0} x} T_{B1} e^{ik_{B1} \cos \theta_{B1} x} + T_{n1} e^{k_{n1} x} \quad (9)$$

where k_B is the bending wavenumber and k_n is the nearfield bending wavenumber.

The bending displacement on plates 2–4 is the sum of a travelling wave and an nearfield wave only. For plate 3, the subscript 1 need only be changed to 3 (and the term T_{B0} is omitted) but for plates 2 and 4 the sign of the wavenumber must also be changed to account for the fact that the waves are travelling in the opposite direction (but still away from the joint).

For in-plane waves the displacement in the x -direction (on plate 1) will consist of components of reflected longitudinal (T_{L1}) and transverse waves (T_{T1}) together with components of the incident wave (T_{L0} and/or T_{T0}) if the incident wave is an in-plane wave to give,

$$\begin{aligned}\xi_1 = & T_{L0} \cos \theta_{L0} e^{-ik_{L0} \cos \theta_{L0} x} + T_{T0} \cos \left(\theta_{T0} + \frac{\pi}{2} \right) e^{-ik_{T0} \cos \theta_{T0} x} \\ & + T_{L1} \cos \theta_{L1} e^{ik_{L1} \cos \theta_{L1} x} + T_{T1} \cos \left(\theta_{T1} + \frac{\pi}{2} \right) e^{ik_{T1} \cos \theta_{T1} x}\end{aligned}\quad (10)$$

The $\pi/2$ terms account for the fact that in transverse waves the particle motion is at right angles to the direction of wave propagation. The displacement on plates 2–4, include only transmitted components of the longitudinal and transverse waves and as with the bending waves the sign of the wavenumber is changed if the waves are travelling in the opposite direction to the incident wave.

The displacement in the y -direction (for plate 1) is similarly given by,

$$\begin{aligned}\zeta_1 = & T_{L0} \sin \theta_{L0} e^{-ik_{L0} \cos \theta_{L0} x} + T_{T0} \sin \left(\theta_{T0} + \frac{\pi}{2} \right) e^{-ik_{T0} \cos \theta_{T0} x} \\ & + T_{L1} \sin \theta_{L1} e^{ik_{L1} \cos \theta_{L1} x} + T_{T1} \sin \left(\theta_{T1} + \frac{\pi}{2} \right) e^{ik_{T1} \cos \theta_{T1} x}\end{aligned}\quad (11)$$

Again the in-plane displacement in plates 2–4, will be composed of only components of the transmitted longitudinal and transverse waves.

When computing the transmission coefficients incident bending, longitudinal and transverse waves are considered in turn and the amplitude of the other two possible incident waves would be set to zero.

The bending, longitudinal and transverse wavenumbers, k_B, k_L and k_T may be given by [5],

$$k_B = \left(\frac{\rho_s \omega^2}{B} \right)^{1/4}, \quad k_L = \left(\frac{\omega^2 (1 - \mu^2) \rho}{E} \right)^{1/2}, \quad k_T = \left(\frac{\omega^2 2(1 + \mu) \rho}{E} \right)^{1/2} \quad (12)$$

and the bending nearfield wavenumber k_n , is given by,

$$k_n^2 = k_B^2 + k_B^2 \sin^2 \theta_B \quad (13)$$

The angles of transmission of any wave on any plate θ_j with wavenumber k_j is related to the angle of incidence of the incident wave θ_0 (with wavenumber k_0) through Snell's law giving,

$$k_0 \sin \theta_0 = k_j \sin \theta_j \quad (14)$$

The angles are taken to be in the range $0 < \theta < \pi/2$.

When considering conditions at the interface between two plates the slope of a plate, ϕ , is equal to $\partial \eta / \partial x$ and the moment, M , and force, F , acting on the boundary are given by [5],

$$\begin{aligned}
 M &= -B \left(\frac{\partial^2 \eta}{\partial x^2} + \mu \frac{\partial^2 \eta}{\partial y^2} \right), F_z = B \left(\frac{\partial^3 \eta}{\partial x^3} + (2 - \mu) \frac{\partial^3 \eta}{\partial y^2 \partial x} \right), \\
 F_x &= -\frac{Eh}{1 - \mu^2} \left(\frac{\partial \xi}{\partial x} + \mu \frac{\partial \zeta}{\partial y} \right), F_y = -\frac{Eh}{2(1 + \mu)} \left(\frac{\partial \xi}{\partial y} + \frac{\partial \zeta}{\partial x} \right)
 \end{aligned}
 \quad (15)$$

2.3. Modelling the frame as a beam

When the element connecting the plates is a beam then continuity requires that the displacement and slope of either side of the beam be the same to give

$$\begin{aligned}
 \eta_1 &= \eta_2, \eta_1 = \eta_3, \eta_1 = \eta_4 \\
 \xi_1 &= \xi_2, \xi_1 = \xi_3 - \phi_1 d, \xi_3 = \xi_4 \\
 \zeta_1 &= \zeta_2, \zeta_1 = \zeta_3, \zeta_1 = \zeta_4 \\
 \phi_1 &= \phi_2, \phi_1 = \phi_3, \phi_1 = \phi_4.
 \end{aligned}
 \quad (16)$$

It is assumed that the plates (1–4) meet at the centre of the beam and that therefore the width of the beam, b , can be ignored. This is reasonable as the nails and screws are nominally located on the centre line of the frame. The depth of the beam, d , is however considered and affects the relationship between the in-plane displacement of plates 1 and 3.

Equilibrium requires that the moments and forces acting on the beam sum to zero. Using the notation given in Fig. 4 gives the equations for the sum of the moments as,

$$M_1 - M_2 + M_3 - M_4 - F_{x1} \frac{d}{2} + F_{x2} \frac{d}{2} + F_{x3} \frac{d}{2} - F_{x4} \frac{d}{2} = M^\diamond \quad (17)$$

The symbol $\diamond$ is used to denote terms relating to the beam (frame). The right hand side term $M^\diamond$ is the resistance of the beam and is given by [3],

$$M^\diamond = -\omega^2 \ominus \phi_1 - \frac{G \partial^2 \phi_1}{\partial y^2} \quad (18)$$

where $\ominus$ is the rotary inertia and is $((b^2 + d^2)/12)bd\rho_i$, G is the torsional stiffness. The terms $F_x d/2$ represent the moment caused by the in-plane plate edge forces acting on the beam. If the finite width of the beam had been included then there would be additional terms $F_z b/2$.

In a similar manner the sum of the forces in the x , y and z direction can be given by,

$$\begin{aligned}
 F_{z1} - F_{z2} + F_{z3} - F_{z4} &= F_z^\diamond, \\
 F_{x1} - F_{x2} + F_{x3} - F_{x4} &= F_x^\diamond, \\
 F_{y1} - F_{y2} + F_{y3} - F_{y4} &= F_y^\diamond
 \end{aligned}
 \quad (19)$$

The terms $F_z^\diamond$, $F_x^\diamond$ and $F_y^\diamond$ are the resisting forces of the beam in the x , y and z direction and are composed of an inertia term and a term relating to bending or compressional stiffness as [3,5],

$$\begin{aligned} F_z^\diamond &= B_z \frac{\partial^4 \eta_1}{\partial y^4} - \omega^2 \rho_l \eta_1, \\ F_x^\diamond &= B_x \frac{\partial^4 (\xi_1 + \phi_1 d/2)}{\partial y^4} - \omega^2 \rho_l (\xi_1 + \phi_1 d/2), \\ F_y^\diamond &= -ES \frac{\partial^2 \zeta}{\partial y^2} - \omega^2 \rho_l \zeta_1 \end{aligned} \quad (20)$$

The first term results from the bending or compression stiffness and the second terms relate to the inertia forces. B_z is the bending stiffness of the beam ($Ebd^3/12$), B_x is bending stiffness of the beam ($Ebd^3/12$) and S is the cross sectional area of the beam. $(\xi_1 + \phi_1 d/2)$ is the beam displacement at its centre expressed in terms of the displacement and slope of plate 1.

If the beam element was massless with zero stiffness then the terms $M^\diamond$ and $F^\diamond$ would be zero.

2.4. Modelling the frame as a plate

An alternative method of modelling the junction is to model the frame as a plate. When the beam is substantial so that d is large but b is still small then the frame will bend along its depth so that ϕ_1 will no longer equal ϕ_3 . This could be included in the beam model by a specific fix to the appropriate equation equating these slopes but is more properly incorporated into the model by modelling the element as a finite length plate. In this case the requirements of continuity and equilibrium must be satisfied at both ends of the plate giving two junctions I and II. The plate configuration used is shown in Fig. 5. The displacements of plates 1–4, are as described in the previous section.

Due to the finite nature of the connecting plate, 5, there will be waves travelling in both a positive and negative direction. The bending displacement on plate 5 can be given by,

$$\eta_5 = T_{B5+} e^{-ik_{B5} \cos \theta_{B5} z} + T_{n5+} e^{-k_{n5} z} + T_{B5-} e^{ik_{B5} \cos \theta_{B5} (z-d)} + T_{n5-} e^{k_{n5} (z-d)} \quad (21)$$

where (+) refers to the amplitude of waves travelling from junction I to junction II, (-) refers to the waves travelling from junction II to junction I. Note that for clarity this is written as junction I being at $z=0$ and junction II being at $z=d$.

The in-plane displacement in the z direction on plate 5 is given by,

$$\begin{aligned} \xi_5 &= T_{L5+} \cos \theta_{L5} e^{-ik_{L5} \cos \theta_{L5} z} + T_{T5} \cos \left(\theta_{T5} + \frac{\pi}{2} \right) e^{-ik_{T5} \cos \theta_{T5} z} \\ &\quad - T_{L5-} \cos \theta_{L5} e^{ik_{L5} \cos \theta_{L5} (z-d)} + T_{T5-} \cos \left(\theta_{T5} + \frac{\pi}{2} \right) e^{ik_{T5} \cos \theta_{T5} (z-d)} \end{aligned} \quad (22)$$

and the in-plane displacement in the y direction may be given by,

$$\begin{aligned}\zeta_5 = & T_{L5+} \sin \theta_{L5} e^{-ik_{L5} \cos \theta_{L5} z} + T_{T5} \sin \left(\theta_{T5} + \frac{\pi}{2} \right) e^{-ik_{T5} \cos \theta_{T5} z} \\ & + T_{L5-} \sin \theta_{L5} e^{ik_{L5} \cos \theta_{L5} (z-d)} + T_{T5-} \sin \left(\theta_{T5} + \frac{\pi}{2} \right) e^{ik_{T5} \cos \theta_{T5} (z-d)}\end{aligned}\quad (23)$$

The boundary conditions for junction I are similar to that described where the frame is modelled as a beam. At junction I where ($x=0$ and $z=0$), continuity and equilibrium requires,

$$\eta_1 = \eta_2, \eta_1 = -\xi_5, \xi_1 = \xi_2, \xi_1 = \eta_5, \zeta_1 = \zeta_2, \zeta_1 = \zeta_5 \quad (24)$$

$$\phi_1 = \phi_2, \phi_1 = \phi_5 \quad (25)$$

$$M_1 - M_2 - M_5 = 0 \quad (26)$$

$$F_{z1} - F_{z2} + F_{z5} = 0, F_{x1} - F_{x2} - F_{x5} = 0, F_{y1} - F_{y2} - F_{y5} = 0 \quad (27)$$

Similarly for junction II, where ($x=0$ and $z=d$), continuity and equilibrium requires,

$$\eta_1 = \eta_4, \eta_3 = -\xi_5, \xi_3 = \xi_4, \xi_3 = \eta_5, \zeta_3 = \zeta_4, \zeta_3 = \zeta_5 \quad (28)$$

$$\phi_3 = \phi_4, \phi_3 = \phi_5 \quad (29)$$

$$M_3 - M_4 + M_5 = 0 \quad (30)$$

$$F_{z3} - F_{z4} - F_{z5} = 0, F_{x3} - F_{x4} + F_{x5} = 0, F_{y3} - F_{y4} + F_{y5} = 0 \quad (31)$$

2.5. Modelling the frame as an SEA subsystem

When the size of the frame depth, d , is small then the phase relationship between the waves at junctions I and II on the plate must be included. However, if d is very large then the plate could be modelled as an SEA plate subsystem. Such a model neglects the phase of the waves as they pass from junction I to II and assumes that the vibration field in the centre plate is diffuse. The two junctions I and II are then modelled as independent ‘T’ junctions.

This model would be expected to be the same as a finite plate model at very high frequencies where d is large compared to the wavelength.

For some timber frames the depth d is normally not much greater than the width b . Therefore at frequencies where there are sufficient wavelengths that fit into the dimension d for the plate to be modelled as a SEA subsystem the plate will be acoustically thick and the bending wave equations based on thin plate bending theory will not be appropriate.

Once the transmission coefficients, τ , have been computed angular averaged transmission coefficient, τ_{av} can be found from [3]

$$\tau_{av} = \int_0^1 \tau(\theta) d\sin(\theta) \quad (32)$$

The coupling loss factor can then be computed, using the angular averaged transmission coefficient, from the equation [9]

$$\eta_{12} = \frac{c_{g1} L_{12} \tau_{12}}{2\pi^2 f S_1} \quad (33)$$

where c_g is the group velocity, L is the common boundary length and S is the surface area.

3. Results

In the previous sections two different theories were presented to describe the coupling between the two plates. The first assumed that the plates were connected at randomly located discrete points and the other that the plates were connected continuously along a line. The question naturally arises as to when each theory should be used.

A simple experiment that shows the behaviour of the plate and the connection was carried out. A double wall was excited by airborne sound and the response of the plate and the junction (on the source side) was measured. Two walls were measured, one had a 100×50 mm frame with screws at 100 mm centres and the other had a 50×50 mm frame with screws at 150 mm. The acceleration level was measured at three positions. The lowest acceleration was on the head of a screw fixing the plasterboard to the frame and this level is used as a reference (see Fig. 6). The highest level was the acceleration far from the frame. It can be seen in Fig. 6 that at very low

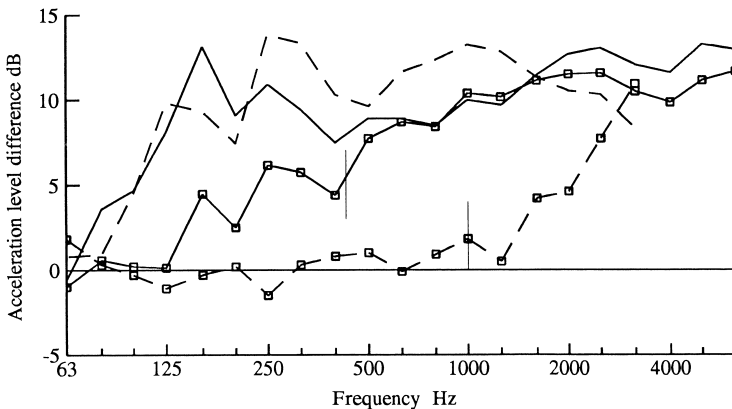


Fig. 6. Acceleration of a wall relative to the acceleration on a screw: —, 100×50 mm frame, 100 mm spacing; - - - - -, 50×50 mm frame, 150 mm spacing; no mark, far from the joint; □, midway between nails; |, estimated transition frequency based on bending wavelength.

frequencies this level is the same as that over the nail. In this frequency region the wall is behaving as a single subsystem and so the vibration should be independent of the position on the wall. Above about 125 Hz the vibration level is about 12 dB higher than on the nail for both walls.

The third measurement was made on the plasterboard mid-way between two nails over the frame. At low frequencies where the wavelength is long (compared to the nail spacing) the vibration between the nails is the same as on the nails. This would be consistent with the junction behaving as a line joint. At high frequencies the vibration between the nails changes to being the same as the centre of the panel. This is consistent with the junction behaving as a series of independent points. The frequency at which the transition occurs is not well defined and is different for the two cases shown in the figure. As the nail spacing increases so the frequency at which the transition occurs increases. A useful approximation for this transition frequency is where half a bending wavelength of bending waves on the plasterboard fits between the nails. For the 100 mm nail spacing this occurs at 430 Hz and for 150 mm spacing at 1000 Hz. This is clearly only an approximation and ignores the properties of the waves on the beam which must also influence the results and which has a much larger bending wavelength.

3.1. Point connection

Tests were carried out on a two plate structure as shown in the insert in Fig. 7 to verify the theory for point connection. This Z-shaped structure was used to ensure that there was no acoustic coupling through the air between the plates. Airborne coupling is important when the structural coupling is weak but is less important when there is stronger structural coupling as occurs with line connection. The equations for the point mobility given earlier do not strictly apply as the mobility at an edge of a plate is not the same as in the centre and in addition moments will occur. However the equations work well in all cases so no correction was made.

The plates were 1.2×2.4 m plasterboard sheets connected to a timber beam 2.4 m in length with the properties given in Table 1. The plates were attached to the timber beam with 3, 7 or 15 steel screws (at approximately 600, 300 and 150 mm centres), of approximately 40 mm in length, to study the effect of varying the number of point connections on sound transmission. The structure was suspended in an anechoic chamber to reduce any possible reflections from surrounding surfaces.

Excitation was by a plastic headed hammer exciting the source plate and the acceleration level difference was measured by placing an accelerometer on each plate. The average acceleration level difference was computed from about 20 positions giving a 95% confidence interval which was less than 1 dB except at some low frequencies.

The internal loss factor of the plasterboard was found from the reverberation time of a freely suspended plasterboard sheet. A standard value of internal loss factor was used for the frame. The total loss factor was computed from the sum of the measured internal loss factor and the predicted coupling loss factors. These procedures were adopted for all tests unless otherwise stated.

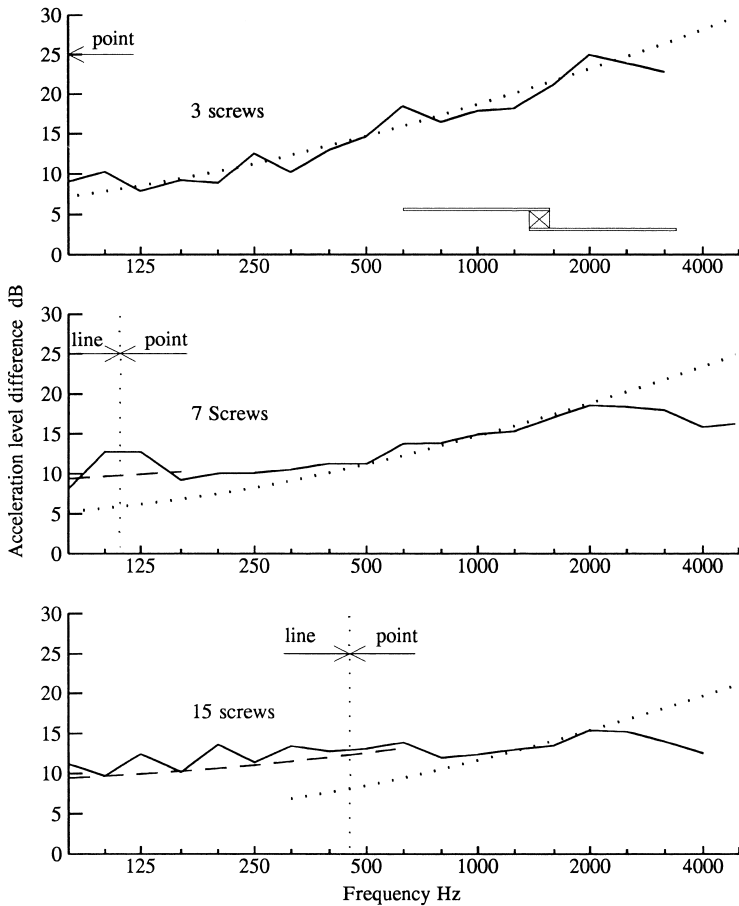


Fig. 7. Measured and predicted acceleration level difference for transmission between two freely suspended plates connected to a 45×75 mm frame by 3, 7 and 15 nails: —, measured; - - - - -, predicted for a line connection;, predicted for a point connection.

The measured and predicted acceleration level difference results are shown in Fig. 7. Over most of the frequency range the point mobility theory applies and it can be seen that there is good agreement between the measured and predicted results up to about 2000 Hz. Above 2000 Hz, all three figures show a decreasing measured level difference.

At low frequencies for the seven and 15 screw cases, the line theory is more appropriate and this is also shown. Again there is good agreement between the measured and predicted results. The frequency at which the transition occurs from line to point coupling was taken to be where a half bending wavelength fits between the screws on the plate and is shown in the figure. There is a small step in the measured results just above the indicated transmission frequency.

Measurements were made on a second test structure with seven screws which was constructed in the same manner except that the screws were placed directly opposite each other so that the coupling is computed by Eq. (4) instead of Eq. (1). The results

Table 1
Material properties of the point connected structure^a

Subsystem	Dimension (m)	Density (kg/m ³)	Young's modulus (×10 ⁹ N/m ²)	Poisson's ratio	Damping
Plate 1	1.17×2.4×0.009	801	2.3	0.2	0.01
Frame	1.2×0.045×0.075	475	8.2	0.3	0.015
Plate 2	1.17×2.4×0.0125	793	2.3	0.2	0.01

^a The values of Young's modulus given are axial values which were used in the calculation of point coupling.

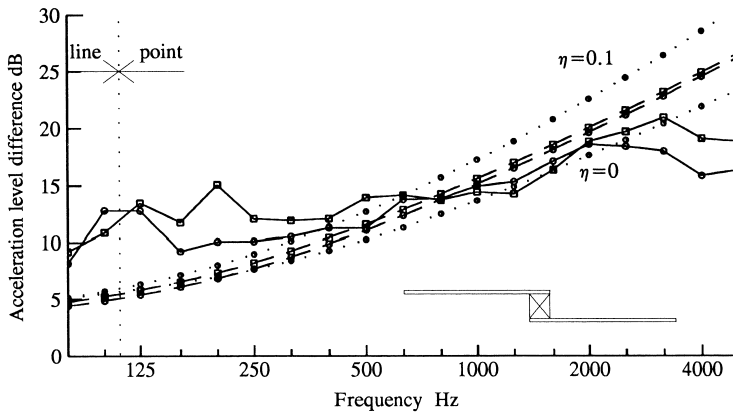


Fig. 8. Measured and predicted level difference between two plates with seven opposite and offset point connections using point connection theory. — □ —, measured offset; - - □ - - -, predicted offset; — ● —, measured opposite; - - ○ - - -, predicted opposite; ··· ··· ● ··· ···, predicted opposite for damping of 0.1 and 0.

for both cases can be seen in Fig. 8. Again the measured level difference decreases above 2000 Hz. It can be seen that there is less than 1 dB between the two predicted results and this is confirmed by the measured results which also show little difference between the results of the two tests.

Since there is little difference between the two theories, the predictions for the real walls, where it is difficult to ascertain whether or not the nails are opposite, were based on the simpler model where the points on opposite sides of the frame were assumed to be independent [using Eq. (1)].

One of the implications of the point model is that transmission between the plates is proportional to the damping of the frame. However, calculations show that the overall level difference is almost independent of the internal damping of the beam. This can be seen in Fig. 8 which shows calculations for two additional cases, one where the beam damping is 0.1 and the other where it is zero. There is little effect on the overall transmission as the total damping of the beam is largely determined by the coupling to the two plates.

If the internal loss factor of the beam is small and the wall is symmetric then half the power going into the beam will go to each plate. Therefore the effective coupling

loss factor for coupling between the two plates will be half the coupling from the plate to the beam. In such a simplified model the beam does not need to be considered as a subsystem though it is used in the calculation of the coupling. This approach would give the level difference shown in Fig. 8 for the case where the damping has been set to zero.

Measurements were made on full size double wall structures with frame depths of 50, 100 and 150 mm. (Additional details are given in Part I). Using the same criteria as before the nails (nominally spaced at 600 mm centres) can be considered to be point connections over the full frequency range. There was no added absorption in the cavity.

Fig. 9 shows the measured and predicted acceleration level difference for the double wall structures. The source was an airborne source to give uniform excitation of the source plate. The predicted model was based on modelling each side of the

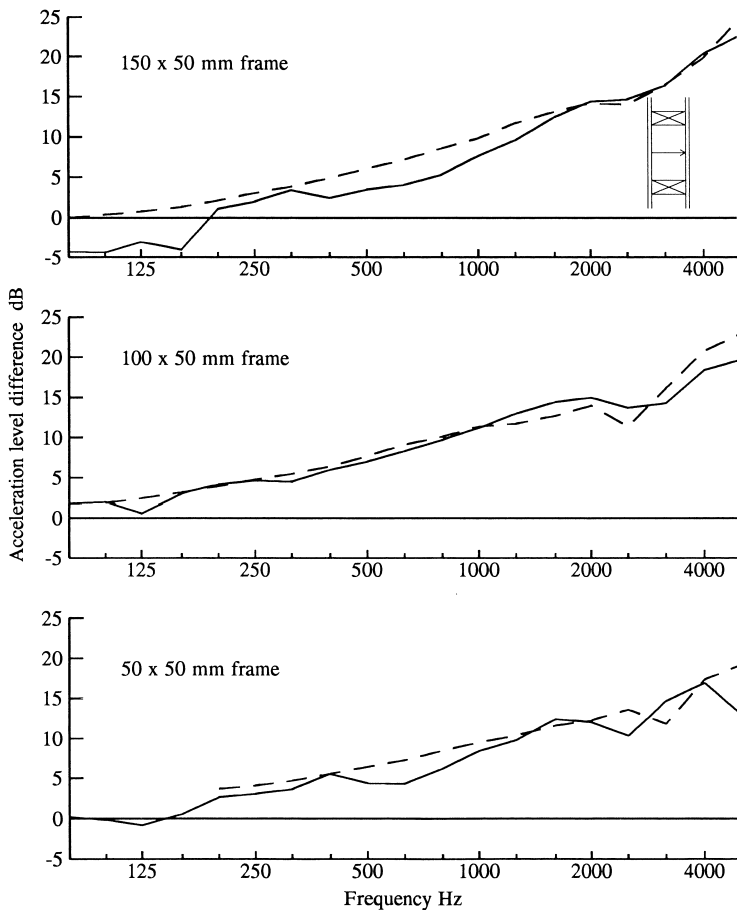


Fig. 9. Measured and predicted acceleration level difference for transmission between two plates that form part of a double wall using an airborne source: —, measured; - - - -, predicted (point connection).

wall as one large plate coupled at random points to the frame's timber studs. It can be seen that there is good agreement between the measured and predicted results for all three walls. The mass–spring–mass resonance for the 50 mm wall is about 100 Hz and this will have affected the low frequency results (leading to negative level difference).

Test were also carried out on the same walls using a plastic-headed hammer as a structural source exciting the plate area between the studs, to generate as uniform a vibration field as possible. The results are shown in Fig. 10 and again there is generally good agreement between the measured and predicted results.

In both Figs. 9 and 10 the dominant sound transmission path between the wall plates is through the structural point connections. This can be established by

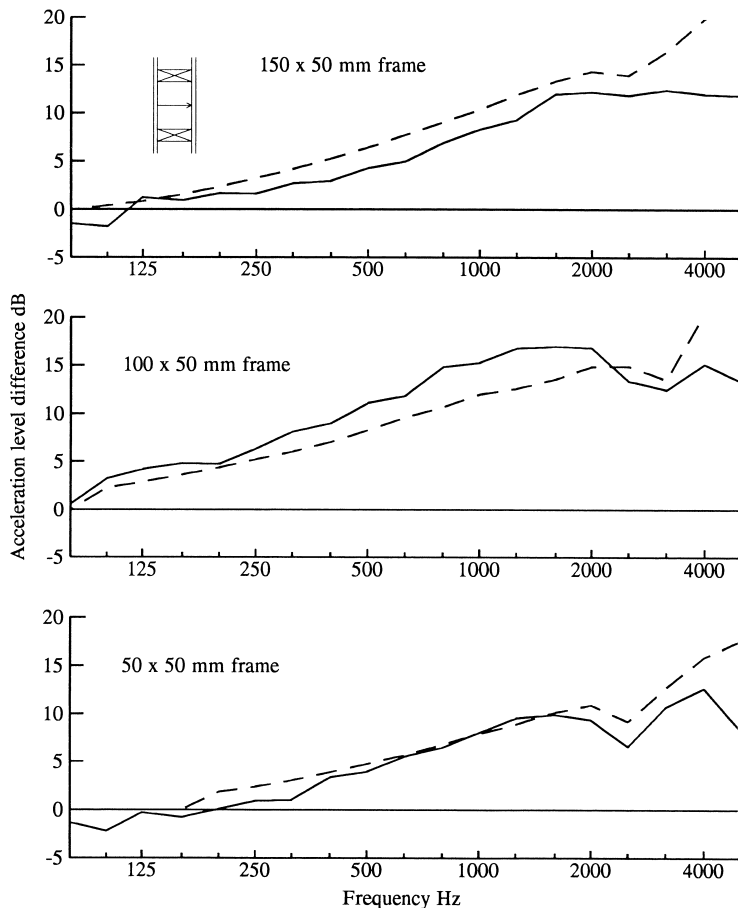


Fig. 10. Measured and predicted acceleration level difference for transmission between two plates that form part of a double wall using a structural source exciting one plate. —, measured; - - - - -, predicted (point connection).

increasing the damping of the cavity and noting whether the predicted structural level difference changes.

If the guidelines of BS5234 [2] are followed and nails are spaced at 150 centres and screws at 300 mm centres then the transition frequencies for 9 mm (12.5 mm) plasterboard with a longitudinal wavespeed of 1800 m/s (approximately Young’s modulus of 2.3×10^9 and density of 800 kg/m³) will be 326 (453) Hz for nails and 82 (113) Hz for screws. Therefore, the point theory will apply over most of the frequency range.

3.2. *Line connection*

When the spacing between the point connections is small it is more appropriate to model the coupling as a line connection. The frame can then be modelled as a beam, a plate or as an SEA subsystem as was described earlier.

A freely suspended ‘H’ structure was tested to compare measured data with the line connection theory. This structure was built with plasterboard plates on either side of a timber beam connected with a row of screws at 60 mm spacings, effectively giving a line connection. This gives a system of four separate plates each 1.2×1.2 m and with a thickness of 12.5 mm, connected to a 1.2 m length timber frame. Details are given in Table 2. There is a considerable difference between the axial, radial and tangential values of Young’s modulus and this results in a frame whose properties not only vary with direction but will vary across its width and depth depending on the way the timber is cut. Axial values give acceleration level differences that were consistently too high whereas radial values (and tangential values) were consistently too low. Therefore in the calculations the geometric mean of the axial and radial values were used. More sophisticated models or more accurate determination of the material properties cannot be justified as the actual values vary from one batten to another and also depends on how the timber is cut.

The measured and predicted results for transmission across a 200×45 mm frame are shown in Fig. 11 for transmission from plates 1 to plates 2–4, using the same measurement procedure as for the point connected plates. Three predicted results are given corresponding to the cases where the frame is modelled as a beam, a plate and an SEA subsystem.

Table 2
Material properties of the point connected structure^a

Subsystem	Dimensions (m)	Density (kg/m ³)	Young’s modulus (×10 ⁹ N/m ²)			Poisson’s ratio	Damping ratio
			Ax.	Rad.	Tan.		
Plates	1.2×1.2×0.0125	806	2.3			0.2	0.01
Frame	1.2×0.0045×0.200	468	8.3	1.1	0.6	0.3	0.015
	1.2×0.045×0.100	419	8.2	1.4	0.9	0.3	0.015
	1.2×0.045×0.050	421	8.2	1.4	0.9	0.3	0.015

^a The geometric mean of the axial and radial values of Young’s modulus were used in the calculation of line coupling.

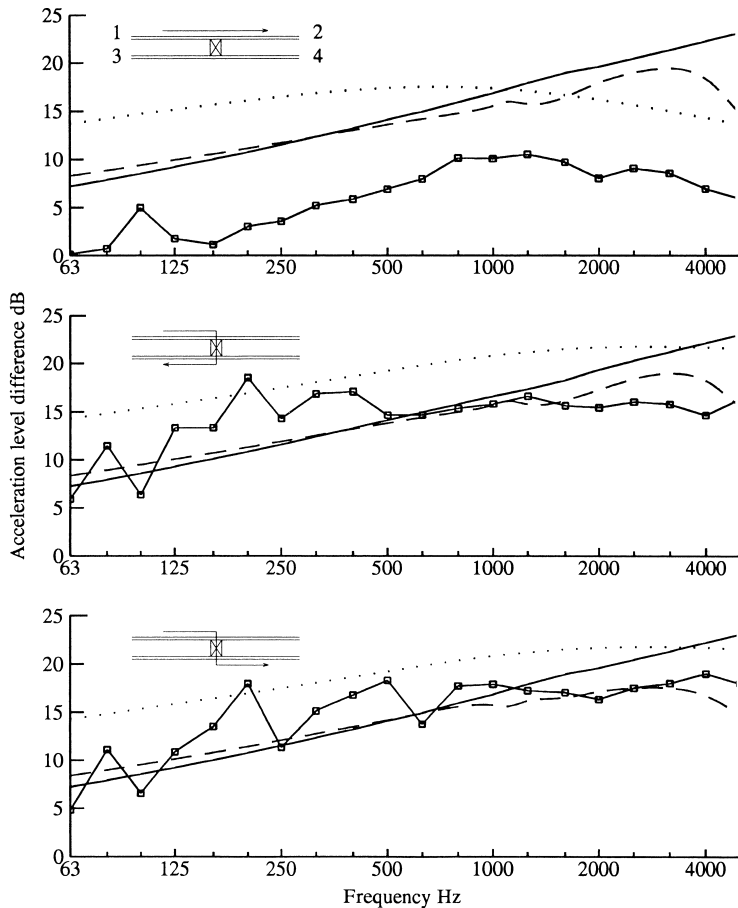


Fig. 11. Measured and predicted acceleration level difference for transmission between two plates that form part of a freely suspended H joint with a 200×45 mm frame. — □ —, measured; —, predicted where the frame is modelled as a beam; - - - - -, predicted where the frame is modelled as a finite plate; ·····, predicted where the frame is an SEA plate subsystem.

The measured results for transmission from plate 1 to 2 are consistently lower than those predicted by all models. Fortunately, for airborne transmission, there is no net power flow between the panels on the same side of a wall and therefore an inability to predict the power flow correctly is not important. The agreement between the measured and predicted results is much better for transmission between plate 1 and plates 3 and 4 with the best agreement being where the frame is modelled as a finite width plate.

When the frame is modelled as a plate then the predicted level difference is not smooth due to fluctuations in the transmission coefficients caused by standing waves in the coupling element which occurs at certain angles of incidence.

Over much of the frequency range the plate and beam models are similar but they do not give the same values even at low frequencies due to the different assumptions

made about the deformation of elements in plate and beam theory. At high frequencies the beam theory predicts a continuing increase in level difference whereas in the plate model the level difference tends to a more constant value. The level difference predicted by both the beam and the plate model would increase if a higher value of Young's modulus (such as the axial value) had been used.

At high frequencies the dimension d is large compared to a wavelength and so the frame can be modelled as an SEA subsystem. Modelling the frame as an SEA subsystem it is only appropriate at very high frequencies where the results tend to those obtained from the plate model.

The effect of varying the depth of the frame on transmission from plate 1 to 4 can be seen in Fig. 12 which shows the results for transmission across a 200, 100 and 50 x 45 mm frame. The test structure was again an 'H' shape structure. The results from

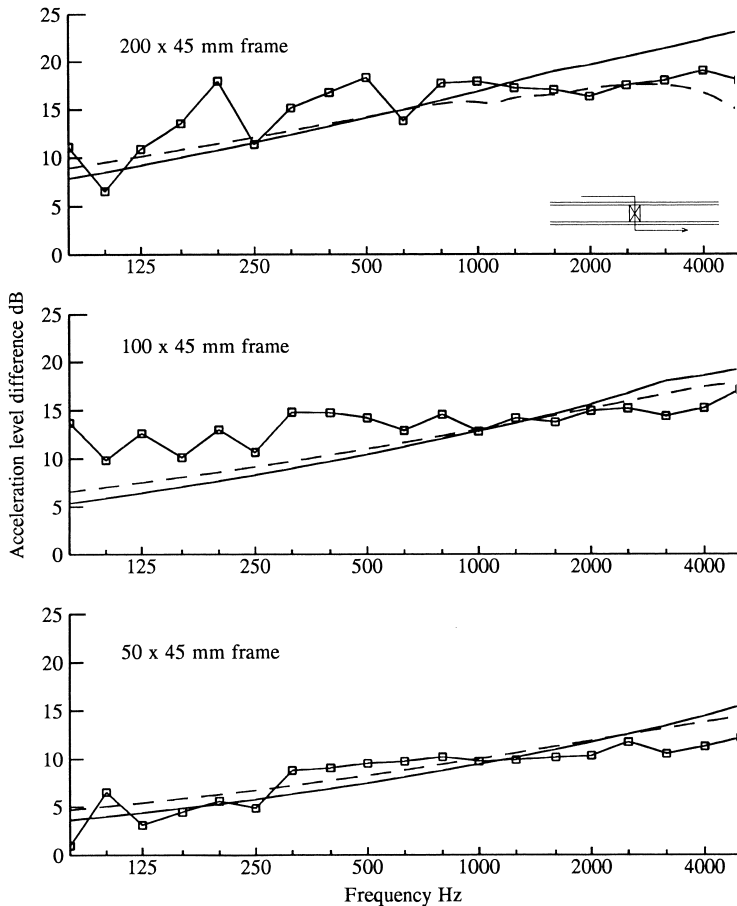


Fig. 12. Measured and predicted acceleration level difference for transmission between two plates that form part of a freely suspended H joint with varying depth: — □ —, measured; —, predicted where the frame is modelled as a beam; - - - - -, predicted where the frame is modelled as a finite plate.

plate 1 to 3 are similar to those for transmission to plate 4. The results show that as the depth increases so the transmission becomes weaker and the acceleration level difference increases by about 3 dB for each doubling of the depth. Overall the agreement between the measured and predicted acceleration level difference is good and again agreement is best with the finite plate model. As the depth increases so there is a greater difference between the beam and plate model results at high frequencies.

Although internal partitions in domestic housing are normally constructed with timber frames, steel channels are also used, particularly in non-load bearing partitions. A test was therefore also carried out on a structure with a steel 'C' channel as the coupling element. This channel is effectively a 70×1 mm steel coupling element and was connected by screws at a nominal 60 mm centres to give a line connection up to 2830 Hz.

The results for the measured and predicted acceleration level difference are shown in Fig. 13. The agreement between the measured and predicted results for transmission from plate 1 to plate 3 and 4 is good. At high frequencies the results for a point model are also shown. A consequence of the point model is that the level difference between plates 1 and 2 should go to zero above 3150 Hz and this does not occur though there may be a trend for the level difference to decrease at higher frequencies.

As with the timber structures the predicted results from plate 1 to 2 under predicts the coupling but accurately predicts the coupling to plates 3 and 4. The results also show that the thin steel plate results in weaker coupling and a slight increase in the level difference particularly at high frequencies.

In addition to the laboratory test structures measurements were made on a real wall with plasterboard attached to a 100×50 mm frame with nails at 30 mm centres. Measurements were made with an airborne source so that all of the sub panels on the source side could be simultaneously excited. A structural source is only reliable when exciting a single panel and as the theory does not accurately predict transmission from one sub-panel to another (plates 1 to 2), so the overall transmission is not well predicted. The results are shown in Fig. 14 and it can be seen that there is excellent agreement between the measured and predicted results. The mass–spring–mass resonance, f_0 , is about 85 Hz for this wall and this will have affected the results at low frequencies. At low frequencies the wall is better modelled as a single subsystem in which case the predicted level difference between the plates will be 0 dB. If the plates act as a mass–spring–mass system then the level difference will be negative up to $\sqrt{2}f_0$. At high frequencies acoustic coupling starts to become important in the predicted results.

4. Discussion

In this paper, two theories for structural coupling between the leaves of a light-weight double leaf partition have been developed. One is based on the assumption that the connection is at independent points and in this theory the frame is included as a subsystem. The other theory is based on a wave theory in which the frame forms a line connection and is not included as a subsystem but as the coupling element.

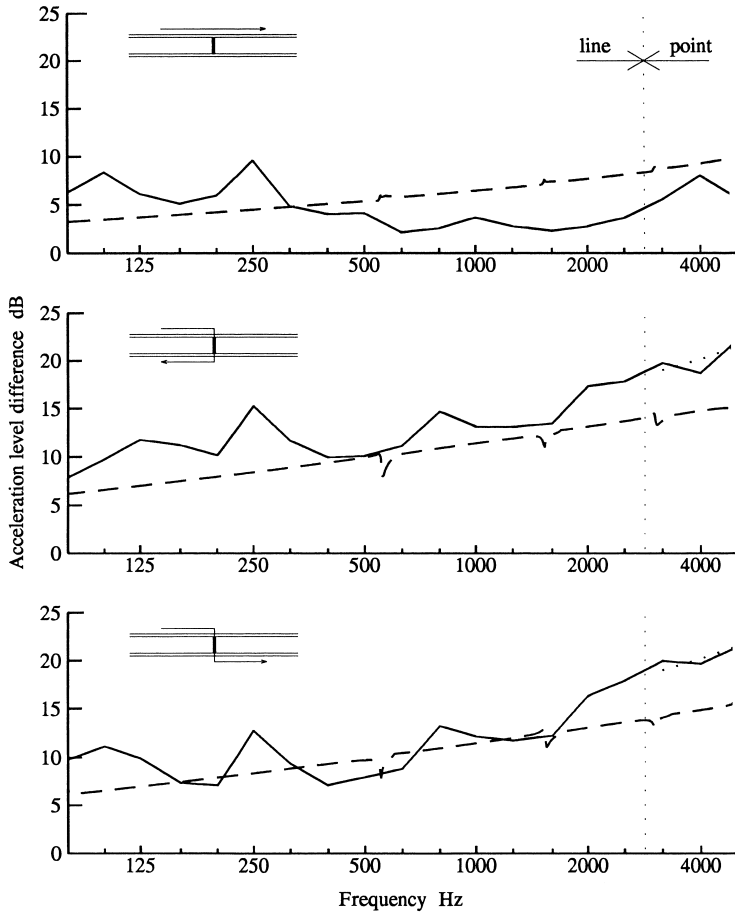


Fig. 13. Measured and predicted acceleration level difference for transmission between two plates connected to a steel frame: ——— measured; - - - - - , predicted where the frame is modelled as a finite plate; , predicted for a point connection.

The point theory is appropriate where the nails or screws are well separated so that they can be considered independent whereas in the line theory it is assumed that the points are so close together that they form a continuous line. The transition point was taken as the frequency where a half-bending wavelength on the plates fitted between the nails or screws. This choice of frequency is somewhat arbitrary but agrees reasonably well with the measured data. However, the wavelength of the frame is much larger and so even when the nails are well separated from the perspective of a wave on the plate they are still close from the perspective of a wave on the frame.

The theory for the line connected plates is more complex and would be appropriate for closely spaced point connections and for glued or welded connections. Two variations of the theory were given. In one the frame was modelled as a beam and in the other it was modelled as a plate. At most frequencies these theories are

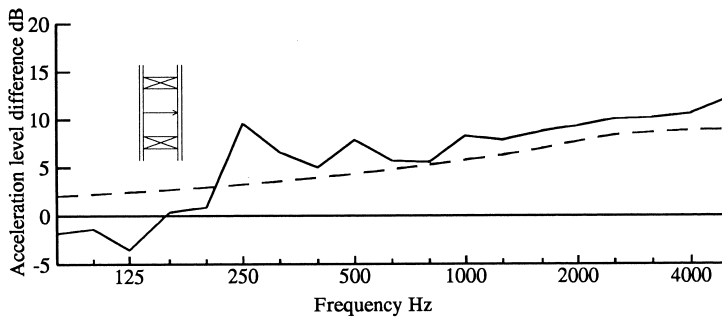


Fig. 14. Measured and predicted acceleration level difference for transmission between two plates connected along a line in a real wall with an acoustic source. — — —, measured; - - - - -, predicted where the frame is modelled as a finite plate.

approximately the same but there are differences at higher frequencies. The theory where the frame is modelled as a plate gives better agreement with the measured data.

Within their respective frequency ranges each of these theories gives good agreement with measured data both for coupling between the two leaves of an isolated test structure and the leaves of real walls. This information can then be included into an SEA model as was shown in Part I of this paper to predict sound transmission through the entire wall.

Acknowledgements

This work was funded by the Engineering and Physical Science Research Council, UK and the Defence Research Agency, Farnborough, UK.

References

- [1] Craik R.J.M., Smith R.S. Sound transmission through lightweight parallel plates, Part I, Airborne sound. *Applied Acoustics* 2000;61(2):223–45.
- [2] BS5234:1991:Code of practice: internal non-loadbearing partitions. London: British Standards Institution, 1992.
- [3] Cremer L, Heckl M, Ungar EE. Structure-borne sound. 2nd Berlin: Springer-Verlag, 1988.
- [4] Craik R.J.M., Wilson R. Sound transmission through masonry cavity walls. *J Sound Vib* 1995;179:79–96.
- [5] Langley RS, Heron KH. Elastic wave transmission through plate/beam junctions. *J Sound Vib* 1990;143:241–53.
- [6] Bhattacharya MC, Mulholland KA, Crocker MJ. Propagation of sound energy by vibration transmission via structural junctions. *J Sound Vib* 1971;18:221–34.
- [7] Craik R.J.M., Wilson R. Sound transmission through parallel plates coupled along a line. *Applied Acoustics* 1997;49:353–72.
- [8] Bosmans I, Vermeir G. Diffuse transmission of structure-borne sound at periodic junctions of semi-infinite plates. *Journal of the Acoustical Society of America* 1997;101:3443–56.
- [9] Craik R.J.M. Sound transmission through buildings using statistical energy analysis. UK: Gower, 1996.